

Computer Science & IT

Discrete Mathematics

Mathematical Logic

Lecture No. 01

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Recap of Previous Lecture



Topic

Distance, Eccentricity, Diameter, Radius and Girth

Topic

Spanning Trees

Topics to be Covered



Topic

Propositions and their types



Topic

Connectives



Topic

Tautology, Contradiction, Contingency and Satisfiable propositional functions



Mathematical Logic

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graph TD; A[Mathematical Logic] --> B[Propositional Logic]; A --> C[Predicate Logic];
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Propositional Logic

Predicate Logic



Topic : Propositional Logic

Proposition : $\rightarrow$

A declarative sentence to which we can assign only one of the truth values (i.e. TRUE/FALSE) is called a propositional statement or proposition.

eg: It will rain today

Cricket match will be played.

India's capital is new delhi

These are all
propositions



Topic : Assumptions of proposition

- ✧ Law of excluded middle : If a proposition is not true then it will be false, and similarly if a proposition is not false then it will be true.
- ✧ Law of contradiction : A proposition can not be true as well as false simultaneously.



Topic : Atomic propositions

Atomic propositions are simple propositions which can not be divided further.

- + It rains today
- + Cricket match will be played.



Topic : Compound Propositions

- Compound propositions are constructed by combining atomic propositions with the help of connectives.

Compound propositions are also known as propositional formula, or propositional function.

eg. It will rain today OR Cricket match will be played
One proposition Connective Another proposition

→ Truth value of Compound Proposition can be determined using Truth values of individual proposition and the Connective being used



Topic : Connectives

Easy

1. Negation ($\neg/\sim$) / Not
2. AND ($\wedge$) / Conjunction
3. OR ($\vee$) / Disjunction

Important

4. Implication ($\rightarrow$)
5. Biconditional ($\leftrightarrow$)



Topic : Negation ($\neg/\sim$)

If p is any proposition, then negation of p / not(p) / $\neg p$ is also a proposition whose truth value is true only when p is false and it is false only when p is true.

p	$\sim p$
T	F
F	T



Topic : AND ($\wedge$) / Conjunction

If 'p' and 'q' are any two propositions, then "p AND q" / $p \wedge q$ is also a proposition whose truth value is true only when both p as well as q are true.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F



Topic : OR ($\vee$) / Disjunction

If p and q are any two propositions, then " p OR q " / $p \vee q$ is also a proposition whose truth value is false only when both p as well as q are false.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F



Topic : Implication ($\rightarrow$)

{ Most important }



↳ "if then"

If p and q are any two propositions, then p implies q (i.e. $p \rightarrow q$ / if p then q) is also a proposition whose truth value is false only when p is true, and q is false.

$p \rightarrow q$
Antecedent Consequent

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

$P \rightarrow Q$ and $\sim P \vee Q$
 have the same truth table
 $\therefore P \rightarrow Q \equiv \sim P \vee Q$

Same truth table
 $\therefore Q \rightarrow P \equiv \sim P \rightarrow \sim Q$

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$	$\sim P \vee Q$	$Q \rightarrow P$	$\sim P \rightarrow \sim Q$	$\sim Q \rightarrow \sim P$
T	T	F	F	T	T	T	T	T
T	F	F	T	F	F	F	F	F
F	T	T	F	T	T	T	T	T
F	F	T	T	T	T	F	T	F

Same truth table, $\therefore P \rightarrow Q \equiv \sim Q \rightarrow \sim P$



Topic : Implication ($\rightarrow$)

- ❑ If p is true, then q must be true for $p \rightarrow q$ to be true.
- ❑ If p is false, then $p \rightarrow q$ is *always* true whatever is the truth value of q .
- ❑ If q is true, the $p \rightarrow q$ is *always* true whatever is the truth value of p .
- ❑ Truth table of $p \rightarrow q$ and truth table of $\neg p \vee q$ are exactly same, therefore both are equivalent.
- ❑ Converse of $p \rightarrow q$ is $q \rightarrow p$.
- ❑ Opposite or Inverse of $p \rightarrow q$ is " $\neg p \rightarrow \neg q$ "
- ❑ Contra positive for $p \rightarrow q$ is " $\neg q \rightarrow \neg p$ "

$$p \rightarrow q \equiv \sim p \vee q$$

$$p \rightarrow q \equiv \underbrace{\sim q \rightarrow \sim p}_{\text{Contra positive of } p \rightarrow q}$$

$$p \rightarrow q \not\equiv \sim p \rightarrow \sim q$$

(inverse of $p \rightarrow q$)

$$p \rightarrow q \not\equiv q \rightarrow p$$

(converse of $p \rightarrow q$)

$$q \rightarrow p \equiv \sim p \rightarrow \sim q$$

Converse of $p \rightarrow q$ is
equivalent to inverse of $p \rightarrow q$



Topic : Biconditional ($\leftrightarrow$)

"if and only if"

If p and q are any two propositions, then " $p \leftrightarrow q$ " (i.e. p *if and only if* q) is also a proposition whose truth value is true only when both p and q have same truth value.

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

We can observe
 $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$



Topic : Biconditional ($\leftrightarrow$)

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$p \leftrightarrow q$ is true if and only if both $p \rightarrow q$ and $q \rightarrow p$ are true.



Topic : Tautology



Tautology is a term
which is defined w.r.t.
Propositional logic

- ★ A propositional function which is always true is called tautology.
- ★ A propositional function which is always true is also called a valid propositional function.

★ From propositional logic point of view
Tautology and Valid propositional function
are same thing

But it will not
be the case w.r.t.
predicate logic

eg.

$$p \vee \sim p$$

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T

} it is always true
∴ $p \vee \sim p$ is a tautology.

eg.

Check if $p \rightarrow (q \rightarrow p)$ is a tautology or not

p	q	$q \rightarrow p$	$p \rightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

} $p \rightarrow (q \rightarrow p)$ is always true

∴ $p \rightarrow (q \rightarrow p)$ is a tautology.

p	q	r	$\dots$	$f(p, q, r, \dots)$
				T
				T
				$\vdots$
				T

If $f(p, q, r, \dots)$ is always true then it is a tautology

$\rightarrow f(p, q, r, \dots)$ is a propositional function over the propositional variables $p, q, r, \dots$



Topic : Contradiction



- A propositional function which is always false is called Contradiction.

➤

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

} Always false
∴ $p \wedge \sim p$ is a Contradiction.

[illegible]

Note:-

① If a propositional function is always true then it is called Valid propositional function

② If propositional function is false for at least one case, then it is an invalid Propositional function

Each Contradiction is invalid propositional function, but Each invalid propositional function need not be Contradiction



Topic : Contingency



- A propositional function which is neither a tautology nor a contradiction is called a contingency.
i.e., A contingency is true for at least one case as well as false for at least one case.

eg:-

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

True for at least one case
as well as

false for at least one case

∴ $p \rightarrow q$ is a Contingency.



Topic : Satisfiable

A propositional function which is true for at least one case is called a satisfiable propositional function.

→ i.e., if a propositional function is "not satisfiable", then it is a Contradiction
↓
not true for at least one case
i.e., false for all cases.

→ If a propositional function is not a Contradiction, then it is a satisfiable propositional function.



Topic : NOTE



6 Every propositional function which is not a contradiction is satisfiable.

- Every tautology is satisfiable but not vice-versa.

f.i.e, Every Satisfiable Propositional function need Not be a tautology

- Every contingency is satisfiable, but every satisfiable function need not be contingency.

eg: Tautologies are also Satisfiable, but they are Not Contingencies.



Topic : NOTE



Tautology	Contradiction	Contingency
✓ Always True	✓ Always False	✓ Neither always true nor always false
✓ Valid	✓ Invalid	✓ Invalid
Satisfiable	Not satisfiable	Satisfiable

→ M.W. :-

→ Read about Equivalences in propositional logic

if possible → Argument & Rules of inference



2 mins Summary



Topic

Propositions and their types

Topic

Connectives

Topic

Tautology, Contradiction, Contingency and Satisfiable propositional functions

THANK - YOU